Laboratory Accuracy Is Not Field Readiness: A Leakage-Controlled Audit of a Review-Gated Crop Disease and Weed Diagnosis System

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Abstract

Plant-disease classifiers often exceed 95% accuracy on curated leaf images, but a diagnosis tool for farmers fails in other ways: photographs reach the wrong crop model, field images differ from training data, and public test sets overlap with training sets. We audited Agri-Doctor Pro, a web application that diagnoses diseases and weeds in maize, rice and cotton through a crop router, crop-specific classifiers, review gates and a vision–language cross-check, by replaying its deployed decision code on every evaluation image. Perceptual hashing showed that 297 of 380 PlantDoc maize photographs (78.2%) copied training images. On the remaining 83, the maize model was 73.5% accurate (95% confidence interval 63.1–81.8%) against 95.0% on laboratory leaves, and 12 of its 22 errors scored at least 0.9. The router sent 334 of 684 rice leaves to the maize model, mostly above its acceptance threshold; a rule that always requests confirmation for maize and rice let none through unconfirmed. Of seven confidence scores, only agreement between whole-leaf and quadrant predictions ranked field errors better than maximum softmax (area under the risk–coverage curve 0.092 vs. 0.164), and it was slightly worse in distribution. Conformal sets kept their coverage on field photographs but averaged five of six labels at 95%, and risk control certified no threshold. A learned gate and few-shot adaptation gave no measurable gain. We release the per-image records and specify the field study that remains.

Index Terms— Conformal prediction, data leakage, dataset shift, plant disease detection, selective classification, weed identification.

I Introduction

Pathogens, pests and weeds remove a large share of crop yield before harvest. Savary et al. [1] estimated global yield losses from pathogens and pests at 30.0% for rice and 22.5% for maize, and an earlier survey by Oerke [2] found that, without crop protection, weeds would cause the largest potential loss of any pest group (34%). Early and correct identification lets a farmer treat the right problem and avoid spraying chemicals the crop does not need. In many farming regions, however, plant pathologists and extension officers are few, and a farmer's first diagnosis is often a visual guess.

Image classifiers seem to offer an easy answer. A network fine-tuned on laboratory leaf images will readily assign a confident label to a field photograph of a maize leaf with northern leaf blight. The confidence score alone says little about whether that label is correct. The image may have been sent to the wrong crop model, it may differ considerably from the training images, and the softmax scores of modern networks are poorly calibrated [3]. Plant-disease models trained on curated collections lose much of their accuracy on images collected under other conditions [4], [5], [6], and predictive uncertainty itself becomes less reliable under dataset shift [7], [8]. A second, quieter problem affects the numbers reported for such models: public plant-image collections are often assembled from overlapping sources and pre-augmented, so test images can be near-copies of training images [9], [10], [11].

Agri-Doctor Pro began as our final-year project. The first version was a Streamlit dashboard that sent a leaf image to a crop-specific model, split the image into quadrants, reported the most frequent disease or weed label among them, and asked a large language model for treatment advice [12]. Tests on photographs from outside the training sets exposed three failures. The crop router assigned rice leaves to the maize model with 92–97% confidence. Quadrant crops of a healthy maize leaf were often classified as common rust, and because the old rule never classified the whole image, one or two such quadrants were enough to produce a disease diagnosis. The advice step then generated a treatment plan for whatever label it received. In each case the user got a detailed answer with no sign that it might be wrong.

An earlier version of this paper described the redesigned system and left most of its decision rules unevaluated. This version reports the evaluation. We re-

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[†]Code, training script and tests: https://github.com/Kireeti2808/agri_doctor (commit ba20fa7). Per-image evaluation records and analysis scripts: see Appendix C.

ran every component of the application's decision logic on held-out and field images, retrained the rice model on a split that keeps near-duplicate images together, compared three backbones over three seeds, and measured how well each candidate confidence signal separates right from wrong answers. Some of the results support the design; several do not, and we report both.

The contributions are as follows.

- A review-gated multi-crop pipeline (Section III) in which each crop classifier predicts diseases and weeds jointly, a crop-review gate stops router suggestions for confusable crops from being applied silently, the whole-leaf prediction is the verdict and quadrant predictions may only corroborate it, and a vision–language model (VLM) that never sees the classifier output provides a second reading that gates treatment advice.

- A leakage-controlled evaluation protocol (Section IV) that replays every image through the deployed decision code, groups near-duplicates by perceptual hash before splitting, and removes field images that duplicate training images. Applied to PlantDoc, it removed 78% of the maize field photographs as copies of training data.

- A validated rice classifier and a backbone study: an EfficientNetV2-B0 rice model at 93.5% held-out accuracy, and a 3×3 comparison (backbones $\times$ seeds) that places ConvNeXt-Tiny, EfficientNetV2-B0 and MobileNetV3-Large on an accuracy–latency curve.

- A selective-prediction study under field shift (Section V): seven confidence scores compared by the area under the risk–coverage curve (AURC) with paired bootstrap intervals, a sweep of the gate thresholds, temperature scaling, split-conformal prediction sets and risk control. Quadrant agreement turned out to be the best error signal on field photographs but not on in-distribution images.

- Negative and null results, reported as such: the router is near chance on rice leaves, the deployed review gate is no better than a softmax threshold at the same coverage, a learned gate and few-shot adaptation give no measurable gain, and the planned VLM measurement failed for an operational reason that we document.

Section II reviews related work and Section III the system. Section IV describes the data, leakage controls and evaluation protocol, and Section V the results. Section VI discusses what they do and do not show, lists threats to validity and limitations, and proposes a field-study protocol; Section VII concludes.

II Related Work

A Leaf-Image Classification and the Laboratory–Field Gap

Most leaf-disease classifiers fine-tune a convolutional network pretrained on ImageNet [13], and a large share train on PlantVillage, a collection of more than 50,000 curated leaf images [14]. Accuracies above 99% [4], [5] and comparisons of fine-tuned architectures [15], [16] come from this setting; surveys cover the wider field [17], [18]. The gap to field images was visible early. Barbedo traced it to the limited size and variety of training sets [19], and the background-pixel result of Noyan [11] is a direct instance of shortcut learning [20]. Field datasets followed: PlantDoc [6], maize northern leaf blight in field imagery [21], maize diseases photographed in Indian experimental fields [22], and rice diseases and pests [23], [24], [25]. SAR-CLD-2024 for cotton [26] and Dhan-Shomadhan for rice [27] are recent field sets for two of our crops. General shift benchmarks show that the gap is not specific to agriculture [8], [28].

B Multi-Crop Systems, Weeds and Patch-Level Analysis

Picón et al. [29] trained one network on five crops and fed the crop identity in as metadata, which raised balanced accuracy on field photographs above both separate per-crop models and a single model without that input. Weed recognition is usually a separate problem, studied on soybean fields [30], rangeland species [31] and cotton weeds [32], and surveyed in [33]. Barbedo classified individual lesions rather than whole leaves and could then detect several diseases on one leaf [34]. Our classifiers place the diseases and weeds of one crop in a single output space, a router chooses the crop, and quadrant analysis is a coarse form of patch-level reading that we use only as a check.

C Mobile Deployment and Vision–Language Models

Compact networks such as MobileNetV2 [35], MobileNetV3 [36], EfficientNet [37] and EfficientNetV2 [38], together with integer quantization [39] and mixed precision [40], make phone inference practical, as shown for cassava [41], [42]. ConvNeXt [43] trades size for accuracy. ImageNet rank predicts transfer rank well [44], but not necessarily rank under shift, which is why we compare backbones on our data. Multimodal models such as Gemini 2.5 [45] describe symptoms without task-specific training, but their agricultural accuracy is uneven. Eight-shot prompting raised the best model's F1 score on plant-stress identification from 46.24% to 73.37% [46], while most open-source models were near chance on weed identification in AgroBench [47]. Success on agronomy exam ques-

tions [48] concerns text knowledge, not diagnosis from a photograph. We therefore use a vision–language model (VLM) only as a second opinion.

D Selective Prediction, Calibration and Conformal Methods

A classifier that may abstain trades coverage for error. Chow derived the optimal reject rule for known posteriors [49]; selective classification was formalized in [50] and given risk guarantees for deep networks in [51]. Maximum softmax probability (MSP) is the standard error-detection baseline [52]; alternatives include ensembles [53], test-time augmentation (TTA) [54] and agreement with a reference classifier [55]. The area under the risk–coverage curve (AURC) summarizes ranking quality over all coverages [56]; calibration is usually reported as expected calibration error (ECE) [57] and corrected by temperature scaling [3]. Conformal prediction gives prediction sets with finite-sample coverage under exchangeability [58], [59], with adaptive variants for image classifiers [60], [61]; risk-controlling prediction sets [62] and conformal risk control [63] extend the guarantee to other losses, including selective error. Exchangeability is precisely what fails between laboratory and field images, so we test these methods on both.

E Leakage and Near-Duplicates

Leakage between training and test data inflates reported performance [9]. Barz and Denzler found near-duplicates of test images in the training sets of CIFAR-10 and CIFAR-100 and measured a 9–14% relative accuracy drop once they were replaced [10]. Plant datasets that ship rotated, flipped or recoloured copies of each photograph as separate files make a file-level split meaningless as a test.

F Positioning

Table I places this work next to representative systems. The comparison is qualitative because datasets and protocols differ. What we add is not a higher accuracy figure but a multi-crop disease-and-weed system with explicit review gates, evaluated under leakage control with statistical tests of each gate.

III System

A Requirements

The failures of the first version translate into five requirements.

R1 *No silent crop errors.* A crop choice known to be unreliable must be confirmed by the user or, if used, shown as unconfirmed.

R2 *Whole-leaf context first.* Partial views must not override a prediction made on the whole image.

R3 *Visible uncertainty.* Weak or inconsistent evidence is shown as a request for review, not hidden behind one label.

R4 *No unsupported advice.* Treatment advice needs more than one source of evidence where possible, carries the system's doubts and stays within safe limits.

R5 *Graceful degradation.* Failure of an external service (model download, weather, VLM) must not stop the core diagnosis.

B Architecture

Figure 1 shows the pipeline. The model-side logic lives in one module without user-interface dependencies (`agri_core.py`, 1,470 lines). It can therefore be unit-tested with mock interpreters, regression-tested with the real TensorFlow Lite (TFLite) files and, for this paper, replayed offline on thousands of labelled images exactly as the application runs it [12]. The Streamlit interface handles input, display and external calls. An uploaded file is verified as an image, its EXIF orientation is applied, images below 64 px on a side are rejected and those above 4096 px are scaled down.

C Crop Routing and the Crop-Review Gate (R1)

A three-class router proposes the crop. Its training code is not in the repository, so we treat it as a given component and evaluate only its behaviour. A suggestion is accepted automatically only when its confidence reaches $\theta = 75\%$ and the crop is not one the router is known to confuse (Algorithm 1). On early regression photographs the router sent three of five rice leaves to the maize model at 92–97% confidence, so maize and rice suggestions are always marked for review and the crop selector starts empty. The classifier-only action cannot run until the user picks a crop; the combined action can run on the suggestion, but the result is flagged and carries a router warning. With the current router this reduces routing to a suggestion for two of three crops; we keep it because cotton is routed without user input and a retrained router can be dropped in unchanged.

D Crop Classifiers and Input Contract

Each crop has one classifier that predicts leaf conditions and weeds together: maize has six classes (northern leaf blight, common rust, gray leaf spot, healthy, broadleaf weed, grass weed), rice eight (bacterial leaf blight, brown spot, healthy, leaf blast, leaf scald, sheath blight and the two weed classes) and cotton ten (five leaf conditions and five weed species). Maize and rice share the two weed

Table I: Representative Plant-Disease and Weed Recognition Studies (Qualitative Comparison)

Study	Crops / task	Images	Weeds	Review / abstain	Field evaluation
Mohanty et al. [4]	14 crops, 26 diseases	PlantVillage (lab)	–	–	small external set
Ferentinos [5]	25 plants	lab and field	–	–	cross-condition test
DeChant et al. [21]	maize, NLB	field	–	–	yes
Picón et al. [29]	5 crops, 17 diseases	phone, field	–	–	yes
Ramcharan et al. [42]	cassava	phone, field	–	–	yes (mobile)
Argüeso et al. [64]	few-shot diseases	field	–	–	yes
Rahman et al. [24]	rice diseases and pests	field	–	–	yes
Haque et al. [22]	maize, 3 diseases	field (India)	–	–	yes
Chen et al. [32]	cotton weeds, 15 species	field	yes	–	yes
This work	maize, rice, cotton; 22 labels	lab + PlantDoc field	yes	crop gate, verdict gate, VLM cross-check, conformal	maize (leakage-screened, $n = 83$); rice and cotton pending

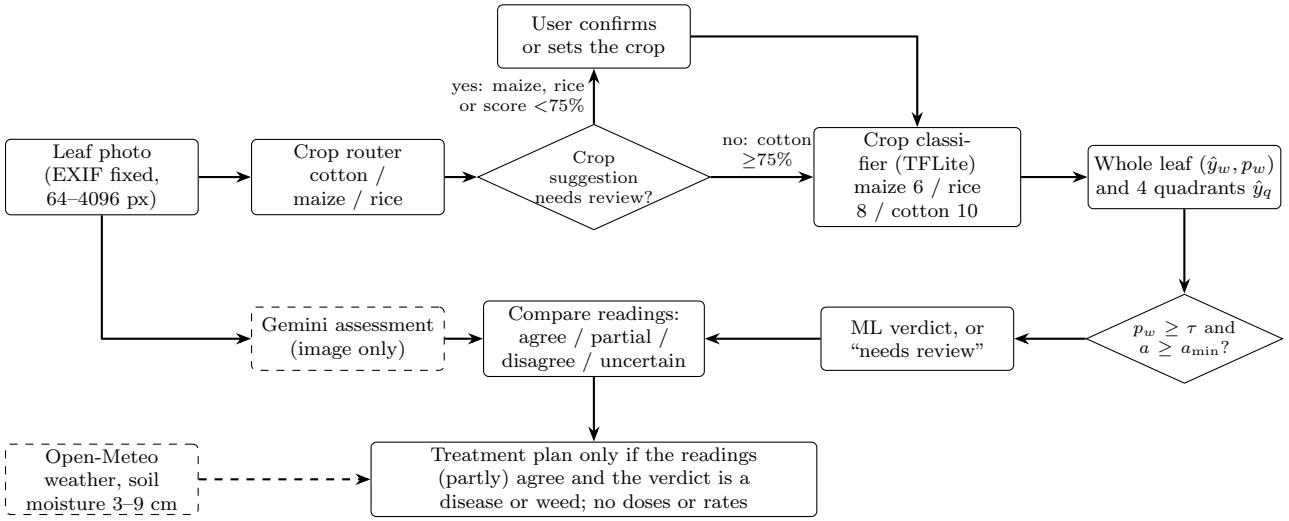


Figure 1: Agri-Doctor Pro pipeline. Diamonds are review gates; dashed boxes are external services. Maize and rice router suggestions are always flagged for confirmation; cotton suggestions at or above 75% are accepted automatically. The combined "Analyze All" action may proceed on an unconfirmed suggestion, but the result is then marked for review. The VLM never receives the classifier output, and the comparison never modifies the ML verdict. The prescription rule shown is that of the combined mode.

labels, giving 22 distinct labels. All models take raw 0–255 RGB images of 224×224 pixels and rescale inside the network; during development, feeding inputs already scaled to $[0, 1]$ made the router predict maize for almost every image, so the contract is enforced in code. A downloaded model is rejected if it lacks the TFLite identifier, is smaller than 100 kB or has the wrong number of outputs, and any prediction whose probabilities do not sum to one within 0.03 raises an error instead of a result.

E Whole-Leaf Verdict with Quadrant Corroboration (R2, R3)

The first version classified the four image quadrants and reported the most frequent disease label above 50%; the whole image was never classified. Quadrants lose the leaf outline and venation, and the maize model tended to call them common rust. In the current design the whole-image prediction is the verdict and the quadrants only check it. Let $\hat{y}_w$ and p_w be the whole-image label and its softmax score and $\hat{y}_q$ the label of quadrant q. The agreement count is

$$a = \sum_{q=1}^{4} \mathbb{I}\left[\hat{y}_q = \hat{y}_w\right], \tag{1}$$

where $\mathbb{I}[\cdot]$ is 1 when its argument holds and 0 otherwise, and the result is marked for review when

$$p_w < \tau \quad \text{or} \quad a < a_{\min}, \tag{2}$$

with $\tau = 0.5$ and $a_{\min} = 2$. Neither value was tuned: the 50% threshold (like the router's 75%) was carried over from the first version, and $a_{\min} = 2$ asks at least half of the quadrants to agree. When $p_w < \tau$ the verdict is reported as uncertain. Non-healthy labels that appear only in quadrants above τ are listed as possible secondary findings and do not change the verdict.

Algorithm 1 Crop routing with a review gate

Require: router probabilities $\mathbf{r}$ over {cotton, maize, rice}; $\theta = 0.75$; confusable set $C = \{\text{maize}, \text{rice}\}$

$c \leftarrow \arg\max \mathbf{r}$; $s \leftarrow \max \mathbf{r}$
if $s < \theta$ **or** $c \in C$ **then**
  show c and all scores; leave crop selector empty
  if user picks c' **then**
    if $c' = c$ **then**
      status $\leftarrow$ CONFIRMED
    else
      status $\leftarrow$ SET MANUALLY
    end if
  else ▷ combined action only
    $c' \leftarrow c$; status $\leftarrow$ NEEDS REVIEW
  end if
else
  $c' \leftarrow c$; status $\leftarrow$ ACCEPTED ▷ cotton only
end if
return c', status

Table II: Rules for Comparing the Classifier and VLM Readings (Checked in Order)

Condition	Category	Plan
ML verdict below τ, VLM certainty low, or VLM label unknown	Uncertain	no
VLM sees another crop or a non-crop plant	Disagreement	no
Same label, or both healthy	Agreement*	yes†
One reading healthy, the other not	Disagreement	no
One finds a disease, the other a weed	Disagreement	no
Same kind of problem, different names	Partial	yes†

*Downgraded to partial if the VLM flags multiple conditions.
†Only when the ML verdict is a disease or weed.

F Blind VLM Cross-Check and Advice Gating (R3, R4)

The combined action makes at most two calls to Gemini (`gemini-2.5-flash`). The first, at temperature 0.2, sends only the photograph (at most 1024 px) and asks for an independent assessment; the classifier result is withheld. A JSON schema restricts the reply to a crop, the closest of the 22 labels (or "other"/"unknown"), observed symptoms and supporting features. It also asks for look-alike explanations such as nutrient deficiency or herbicide injury, image-quality problems, a multiple-conditions flag and a self-reported certainty; values outside the schema become "unknown" or "low". The two readings are compared by fixed rules (Table II); confidences are never averaged, and agreement is presented as support, not confirmation. The second call, which drafts a management plan from weather, modelled soil moisture [65] and both readings, is made only on agreement or partial agreement and only for a disease or weed verdict.

Blindness to the classifier output does not make the two readings independent. A VLM trained on web data may have seen the same public images, and for weeds, where VLMs are close to chance [47], the cross-check probably acts as a noisy veto. The classifier-only action can still request a plan for a disease or weed verdict without a VLM reading; the prompt then carries the review reasons and must say that the diagnosis is uncertain. Withholding plans for flagged results in this mode too is a simple change we recommend (Section C).

G Safety of Generated Advice, Failure Handling and Testing (R4, R5)

The prescription prompt forbids presenting the classification as a confirmed diagnosis, giving pesticide doses, mixing or application rates, or promising recovery. It asks for treatment types generally supported for the crop and condition and tells the user to have an extension officer verify the diagnosis before any chemical treatment (Appendix B). These are instructions to a language model, not guarantees, and no agronomist has yet reviewed a sample of plans. If the API key is missing, the quota or rate limit is reached, the model is unavailable or the network fails, the ML diagnosis is still shown with the reason. Successful VLM responses are cached by image hash and model; failed calls are not cached, so transient errors can be retried. The repository holds 107 test functions in nine files covering intake, routing, quadrant logic, weather parsing, VLM response handling and the training script; tests that need real model files run only when the files are present, and none uses the network.

IV Audit Protocol

A Data

Table III lists the data. *Cotton:* original images of a Kaggle cotton-leaf collection [66] (its augmented copies excluded) plus the five largest classes of CottonWeedID15 [32]; 5,264 images split 70/15/15 by class with no file in two splits. *Maize:* a Kaggle collection compiled from PlantVillage and PlantDoc images [67] (4,188 images, four classes), 200 broadleaf-weed images and the grass class of a soybean weed dataset [30]; 7,908 images, of which a fixed 20% (1,580) was used for validation across several training runs. *Rice:* six leaf classes of a Kaggle collection [68] (628–653 images each) and the broadleaf (1,191) and grass (3,520) weed classes of [30]; 8,540 images with an imbalance ratio of 5.61. The rice collection is pre-augmented: rotated, flipped and recoloured copies (`aug_*`) sit next to fewer camera originals (`IMG_*`).

Table III: Data Used in This Study

Set	Source	Images	Role
Cotton	[66] + [32]	5,264	train/val/test
Maize	[67] + [30]	7,908	train/val
Rice	[68] + [30]	8,540	train/val/test
Maize field	PlantDoc [6]	83[a]	field test
Cotton field	SAR-CLD-2024 [26]	0[b]	planned
Rice field	Dhan-Shomadhan [27]	0[b]	planned

[a] 380 found; 297 removed as near-duplicates of training images.

[b] Automatic download failed in this run.

Field photographs. For maize we used the corn images of PlantDoc [6], collected from the web and showing leaves in natural settings. The notebook was configured to use SAR-CLD-2024 [26] for cotton and the field-background images of Dhan-Shomadhan [27] for rice; both automatic downloads failed in this run, so neither crop has field results (Section E).

B Leakage Control

Group-aware splitting. The rice training script hashes every image perceptually and links images whose hashes differ by at most four bits. Connected images form a duplicate group, and whole groups are assigned to the training, validation or test split. Because group assignment depends on the seed, the test set changes between seeds (1,391, 1,285 and 2,682 images for seeds 42, 7 and 2024).

Field-set screening. Every field photograph is hashed and compared with the training images of the same crop under the same threshold, and matches are removed. This matters for maize because the laboratory collection already contains PlantDoc images [67].

A perceptual hash catches resized, re-encoded and lightly edited copies, not rotations, crops or strong colour changes, so some augmented siblings may survive a split; Section D examines the consequence.

C Models

All three crop models use an ImageNet-pretrained EfficientNetV2-B0 [38]; Table IV compares the configurations as they appear in the repository and notebooks. The cotton and maize heads add global average pooling, batch normalization, a 256-unit ReLU layer and dropout of 0.3 before the softmax layer; the rice script uses global average pooling, dropout of 0.3 and an L2-regularized softmax layer. The rice model was retrained with `training/train_rice.py` in two phases. The first trains the new head on a frozen backbone (Adam [69], learning rate 10^{-3}, up to 15 epochs); the second unfreezes the top 40 backbone layers with batch normalization held fixed (learning rate 10^{-5}, up to 20 epochs). Both phases use inverse-frequency class weights and early stopping on validation loss (patience 5, best weights restored), and we ran three seeds (42, 7, 2024).

For the backbone study we trained MobileNetV3-Large [36], EfficientNetV2-B0 and ConvNeXt-Tiny [43] with the evaluation notebook's shared head and schedule under the `mixed_float16` policy [40], three seeds each, all on the seed-42 split; holding the split fixed isolates the effect of backbone and initialization. Latency is the median time for one image on the Colab CPU, measured on a CPU copy of each model under the same precision policy; it is a relative figure, not a phone measurement.

D Replay of the Deployed Decision Code

Each evaluation image passes through the application's own functions at commit ba20fa7: router, whole-leaf classification, quadrant classification and agreement count (1), verdict gate (2), the old quadrant-vote rule, and classification of horizontally and vertically flipped copies. One record per image stores the true label, all class probabilities, both flips, the agreement count, the review flag, the old rule's prediction, and the router's crop and confidence, so any score or threshold can be evaluated afterwards without rerunning a model. Router accuracy is measured only on crop-leaf images, since a weed photograph has no single correct crop.

E Confidence Scores

With $\mathbf{p}$ the whole-image probability vector and $p_{(1)} \geq p_{(2)}$ its two largest entries, we rank verdicts by seven scores:

1. MSP, $p_{(1)}$ [52];
2. margin, $p_{(1)} - p_{(2)}$;
3. negative entropy, $\sum_k p_k \log p_k$;
4. flip-TTA confidence, the largest entry of the mean of the original and two flipped probability vectors [54];
5. TTA agreement, the number of flipped copies predicting $\hat{y}_w$;
6. quadrant agreement, a from (1);
7. gate pass, whether (2) passes.

Scores 5–7 are lexicographic: the discrete signal decides first and MSP orders verdicts within a level.

F Statistics

Accuracies and recalls carry 95% Wilson intervals [70]. Two decision rules on the same images are compared with the exact McNemar test [71]. For a score, the risk–coverage curve gives the error rate among the k most confident verdicts for $k = 1, \ldots, n$, and the AURC is its mean [56]; lower is better. Differences between two scores are tested with a paired bootstrap [72] over images (2,000 resamples, percentile intervals, seed 0). The

Table IV: Training Configurations

	Cotton	Maize	Rice (repository script)	Rice backbone study
Backbone	EfficientNetV2-B0	EfficientNetV2-B0	EfficientNetV2-B0	MobileNetV3-L / EffNetV2-B0 / ConvNeXt-T
Head	GAP, BN, Dense 256, dropout 0.3	GAP, BN, Dense 256, dropout 0.3	GAP, dropout 0.3, L2 softmax	shared notebook head
Trainable	last 20 layers, BN frozen	last 20 layers	head, then top 40 layers	shared schedule
Optimizer	Adam, 10^{-4}, batch 32	Adam, 10^{-3}, batch 32	Adam, 10^{-3} then 10^{-5}, batch 32	shared, mixed precision
Class weights	inverse frequency	none	inverse frequency	–
Augmentation	flips, rotation, zoom, contrast	rotation, shift, shear, zoom, h-flip	flips, rotation, zoom, shift, contrast, brightness	–
Split	stratified 70/15/15	fixed 20% validation	duplicate groups, per seed	duplicate groups, seed 42
Selection	lowest val. loss (epoch 19)	last epoch (15)	early stopping, best weights	per notebook
Seeds	1	1	42, 7, 2024	42, 7, 2024 per backbone
TFLite export	dynamic-range quant., 7.0 MB	float, 24.7 MB	float, 23.5 MB	not exported

deployed gate is compared with an MSP threshold that answers the same number of images. Calibration is 15-bin ECE [57] before and after temperature scaling [3], fitted on one half of the images and applied to the other.

Learned gate. A logistic regression combines the confidence features into the probability that a verdict is correct; cross-validation ensures every image is scored by a model that did not see it. *Risk control.* For a target selective error α and failure probability $\delta = 0.1$, a score threshold is chosen on a random calibration half so that, with probability at least $1 - \delta$, selective error on new exchangeable data is at most α [62], [63]. We repeat over 20 random splits and report mean coverage, mean test risk and the number of violations; about $\delta \cdot 20 = 2$ are expected. *Prediction sets.* Split-conformal sets [58], [59] at 90% and 95% nominal coverage contain every class whose probability clears a threshold calibrated on half of the images, again over 20 splits. Sets may be empty, so mean sizes below one are possible. *Few-shot adaptation.* Following the idea of [64] at a small scale, we trained a linear head on frozen ImageNet EfficientNetV2-B0 embeddings of 900 laboratory maize images from the three disease classes in the field set. To these we added 0, 5, 10 or 20 field photographs per class, tested on the remaining field photographs, and drew five random samples per budget. *VLM agreement.* Sixty field photographs per crop were to be sent through the application's assessment function and the agreement categories tabulated against correctness.

The evaluation ran in one Colab notebook on an NVIDIA T4 GPU (TensorFlow 2.21.0, Python 3.13), finishing on 19 October 2026. We recomputed every headline statistic independently from the per-image records with a separate script (Appendix C).

V Results

A Leakage and the Laboratory–Field Gap

Leakage changes the answer. Of the 380 PlantDoc corn photographs, 297 (78.2%) matched a maize training image, leaving 83 field photographs: 51 blight, 20 common rust and 12 gray leaf spot (Table V). Without screening, most of the "field" test set would have been training data. In the rice collection a third of all images (2,821 of 8,540) duplicated another image. The hash check also raised possible label conflicts: at least 50 groups, 941 images in total, contain images filed under different classes. Groups spanning six or seven classes are more likely hash collisions between visually uniform images than true duplicates; we have not audited them by hand, and the rice estimates below include them.

Table V: Near-Duplicate and Leakage Audit

Quantity	Value
PlantDoc corn photographs found	380
removed as copies of maize training images	297 (78.2%)
retained as field test set	83
Rice images	8,540
duplicate groups (Hamming ≤ 4)	398
images beyond the first in a group	2,821 (33.0%)
groups with mixed labels	≥ 50 (941 images)
Rice test split, seed 42	1,391
camera originals (`IMG_*`)	40
pre-augmented copies (`aug_*`)	568
other files (incl. weed images)	783

Laboratory performance. On 790 held-out cotton images the cotton model was 95.57% accurate (93.9–96.8%), with macro precision, recall and F1 of 0.956, 0.963 and 0.959; the 7.0 MB TFLite export scored 95.82%, two images more. All 35 errors stayed within a group—leaf conditions were confused only with leaf conditions and weeds only with weeds—and accuracy was 97.77% on leaves (95.2–99.0%) and 94.43% on weeds (92.1–96.1%). The most frequent error was carpetweed predicted as waterhemp (7 of 115). The maize model reached 97.28% (96.4–98.0%) on its 1,580-image validation split. Grass-weed images from a different dataset, however, form 44.6% of that split and were all correct, so we use the 836 maize leaves as the reference: 94.98% (93.3–96.3%), macro F1 0.935. This validation split is a file-level split of a collection that includes PlantDoc images and was reused across training runs, so even 94.98% may be optimistic.

The field gap. On the 83 screened field photographs the maize model was right 61 times: 73.49% (63.1–81.8%), 21.5 points below the laboratory figure and far outside both intervals. Figure 2 and Fig. 3(b) show where the loss comes from. Common rust held up (18 of 20, 90.0%). Blight fell to 76.5% (39 of 51), with seven photographs called common rust and four called a weed. Gray leaf spot collapsed to 4 of 12 (33.3%, 13.8–60.9%), with five called common rust and three blight. The drift toward

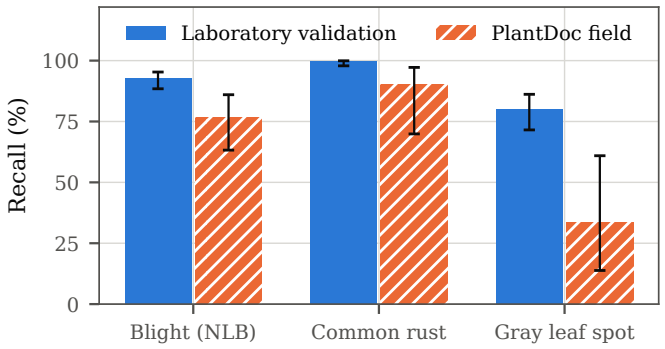


Figure 2: Per-class recall of the maize model on its laboratory validation split and on the 83 leakage-screened PlantDoc field photographs, with 95% Wilson intervals.

Table VI: Crop Router and Crop-Review Gate

	Maize field	Rice leaves (test)
Images	83	684
Routed to correct crop	78 (94.0%)	350 (51.2%)
Routed to maize / rice / cotton	78 / 3 / 2	334 / 350 / 0
Misrouted with confidence $\geq 75\%$	2	248 (74.3%)
Wrong crop auto-accepted	2 (2.4%)	0 (0.0%)

common rust first seen on quadrant crops reappears here on whole photographs.

Field errors are confident. Twelve of the 22 field errors (55%) had a softmax score of at least 0.9, and their median was 0.916. On the in-distribution rice test only 2 of 91 errors (2.2%) reached 0.9, with a median of 0.564. A softmax threshold cannot catch errors in this regime, which is what studies of dataset shift predict [7].

B Routing and the Crop-Review Gate

Table VI summarizes the router. On maize field photographs it chose maize 78 times out of 83 (94.0%, 86.7–97.4%). On the 684 rice leaves of the rice test split it chose rice only 350 times, 51.2% (47.4–54.9%), close to a coin flip between rice and maize; it never chose cotton. Healthy rice leaves fared worst, with 124 of 131 sent to maize; 59 of 96 sheath-blight leaves followed, and brown spot was misrouted least often (30 of 124).

The misrouted leaves were not low-confidence cases (Fig. 4): their median router confidence was 87.9%, and 248 of the 334 (74.3%) were at or above the 75% threshold. A confidence-only gate would have let three quarters of them be diagnosed silently by the maize model. The crop-review gate, which flags every maize and rice suggestion whatever its confidence, let none through: 0 of 684 rice leaves were auto-accepted with the wrong crop (upper 95% bound 0.6%). This is the clearest result in the paper in favour of the design, and it rests on a rule introduced after a handful of regression photographs.

The gate is not complete. Two maize field photographs were routed to cotton with 85.3% and 96.6% confidence and, because cotton suggestions at or above 75% are accepted automatically, would have been diagnosed by the cotton model without warning: 2 of 83, 2.4% (0.7–8.4%). The three photographs routed to rice (51–72% confidence) were flagged.

Table VII: AURC of Seven Confidence Scores (Lower Is Better; 95% Bootstrap Interval)

Score	Maize field ($n = 83$)	Rice test ($n = 1{,}391$)
Max softmax	0.164 (0.084–0.275)	0.0055 (0.0038–0.0074)
Margin	0.165 (0.084–0.276)	0.0058 (0.0040–0.0077)
Neg. entropy	0.163 (0.081–0.274)	0.0055 (0.0038–0.0075)
Flip-TTA confidence	0.142 (0.070–0.238)	0.0052 (0.0036–0.0071)
TTA agreement + conf.	0.157 (0.079–0.269)	0.0056 (0.0039–0.0075)
Quadrant agreement + conf.	**0.092** (0.044–0.159)	0.0076 (0.0053–0.0102)
Gate pass + conf.	0.136 (0.069–0.233)	0.0061 (0.0041–0.0083)
Learned gate (CV)	0.128	0.0054
Error at full coverage	0.265	0.0654

C Which Signals Flag Errors?

Whole-leaf verdict versus quadrant vote. Replacing the quadrant vote by the whole-leaf verdict (R2) was the main change to the decision rule. On the rice test split the whole-leaf verdict was right on 93.46% of images and the old vote on 85.41%; 144 images were right only under the new rule and 32 only under the old (exact McNemar $p = 3.6 \times 10^{-18}$). With the currently deployed rice model the comparison is 98.71% against 87.20% (168 against 8). On maize field photographs the two rules tied at 73.49%, with eight discordant images each way ($p = 1.0$): much better in distribution, no worse—and no better—under shift.

Ranking quality. Table VII gives the AURC of the seven scores and Fig. 5 the risk–coverage curves of three of them. In distribution the softmax-based scores are nearly interchangeable (0.0052–0.0058), and quadrant agreement is worse than MSP: 0.0076 against 0.0055, a paired difference of +0.0022 (95% interval +0.0009 to +0.0035). Under field shift the order reverses. Quadrant agreement has an AURC of 0.092 against 0.164 for MSP, a difference of -0.072 (-0.151 to -0.012), and it is the only score whose interval excludes zero. The bootstrap one-sided tail is 0.003, so the two-sided $p \approx 0.006$ survives a Bonferroni correction for six comparisons ($0.05/6 \approx 0.008$). The gain is concentrated at low and medium coverage: answering the half of the photographs with the highest agreement gives a 4.8% error rate, against 14.3% for MSP. With $n = 83$ and 22 errors the effect size is uncertain, and the set of seven scores, though chosen before the experiment, was not pre-registered.

Figure 6 shows why. On field photographs, 27% of wrong verdicts (6 of 22) had fewer than two agreeing quadrants against 3% of correct ones (2 of 61), and only one wrong verdict had all four quadrants agreeing. On the rice test split the count is also informative—66% of errors (60 of 91) but 11% of correct verdicts had $a < 2$,

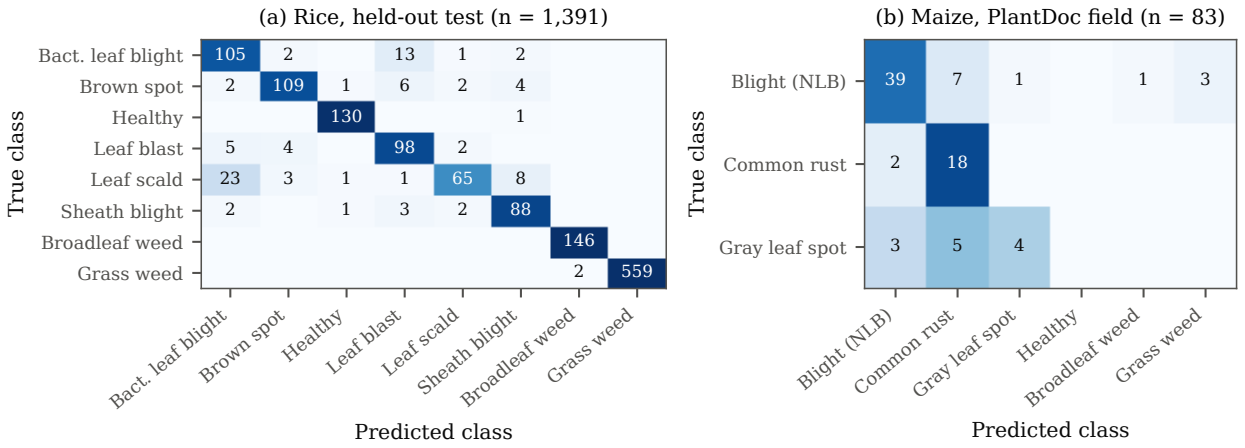


Figure 3: Confusion matrices replayed through the application code. (a) Retrained rice model (repository script, seed 42), not the currently deployed rice model, on the held-out test split. (b) Maize model on the PlantDoc field photographs; rows are the three disease classes present, columns all six outputs. Counts; shading is row-normalized recall.

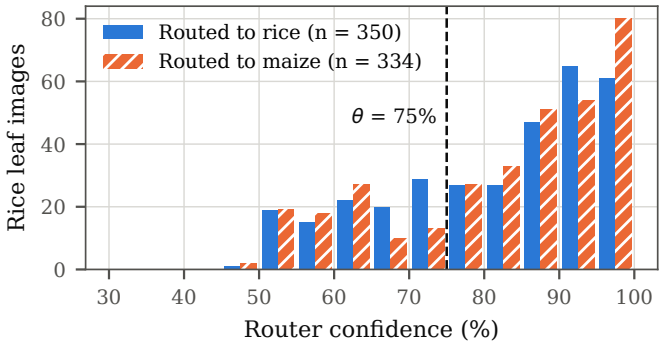


Figure 4: Router confidence on the 684 rice leaves of the test split, by routing decision. Most leaves sent to the maize model lie above the 75% acceptance threshold (dashed), so only the crop-based review rule stops them.

Table VIII: Deployed Verdict Gate ($\tau = 0.5$, $a_{\min} = 2$) vs. Max Softmax at Matched Coverage

	Maize field	Rice test
Flagged for review	9 / 83 (10.8%)	220 / 1,391 (15.8%)
Errors caught	6 / 22 (27.3%)	69 / 91 (75.8%)
Correct verdicts flagged	3 / 61 (4.9%)	151 / 1,300 (11.6%)
Accuracy, unflagged	78.38%	98.12%
Max softmax, same coverage	75.68%	99.06%
Difference (95% CI), points	+2.7 (−1.5, +8.2)	−0.94 (−1.56, −0.25)

Table IX: Expected Calibration Error (15 Bins) Before and After Temperature Scaling

Set	Raw	Scaled	Temperatures (2 folds)
Rice test	0.028	0.009	0.86, 0.75
Maize field	0.235	0.192*	4.36, 4.12

*Range 0.125–0.200 over 20 random two-fold splits.

and no verdict with $a = 4$ was wrong—but it ranks errors less finely than the softmax score, which in distribution already orders them well.

The deployed gate and its thresholds. The gate (2) flagged 15.8% of rice test images and 10.8% of maize field photographs (Table VIII). On rice it caught 69 of 91 errors (75.8%) at the cost of flagging 151 correct verdicts, and accuracy on unflagged images was 98.12%. An MSP threshold answering the same 1,171 images reached 99.06%, so at matched coverage the gate is 0.94 points worse (−1.56 to −0.25). On the field photographs it caught only 6 of 22 errors and was 2.7 points better than MSP at matched coverage, an interval (−1.5 to +8.2) that includes zero. Sweeping τ for each $a_{\min}$ (Fig. 7) shows the same split. On maize field photographs, requiring $a \geq 3$ answers 68.7% of images at 87.7% accuracy against 84.2% for MSP at the same coverage, and requiring $a = 4$ answers 41% at 97.1% against 85.3%. On rice, requiring $a \geq 2$ or $a \geq 3$ costs 1.3 and 1.1 points against MSP at equal coverage. The thresholds that are best in distribution and under shift differ, and choosing them needs a field calibration set.

Calibration. The two settings miscalibrate in opposite directions (Table IX, Fig. 8). On rice the model was slightly underconfident: the fitted temperatures were 0.86 and 0.75, and scaling cut ECE from 0.028 to 0.009. On maize field photographs it was strongly overconfident, with ECE 0.235 and temperatures of 4.4 and 4.1. Scaling reduced field ECE to 0.192 in the notebook's split; over 20 further random two-fold splits the scaled ECE ranged from 0.125 to 0.200 (median 0.157), which says as much about ECE on 83 images as about the remedy. A temperature fitted on laboratory images would sharpen field predictions that are already overconfident, so calibration must be fitted on field data.

Learned gate, risk control and prediction sets. The

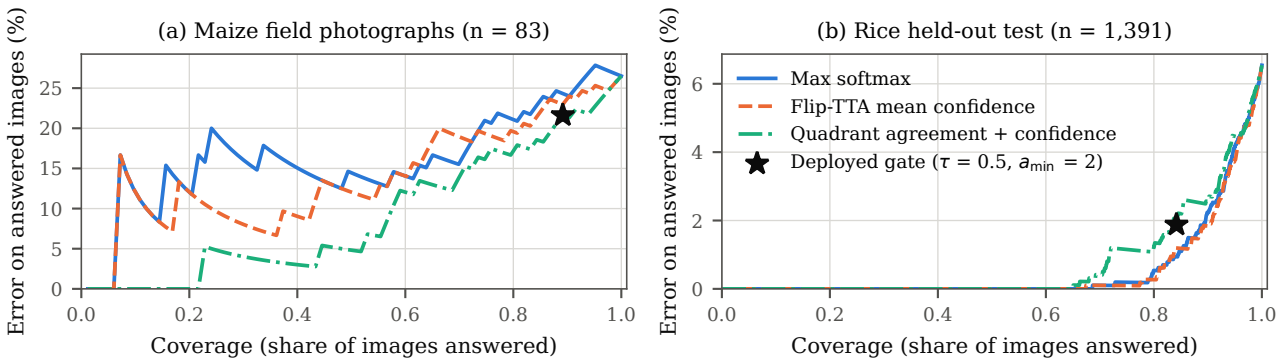


Figure 5: Risk–coverage curves: error among the images answered when only the most confident fraction is answered. The star marks the deployed gate ($\tau = 0.5$, $a_{\min} = 2$). (a) On maize field photographs quadrant agreement gives the lowest error at most coverages. (b) On the rice test split the softmax-based scores are best and nearly identical.

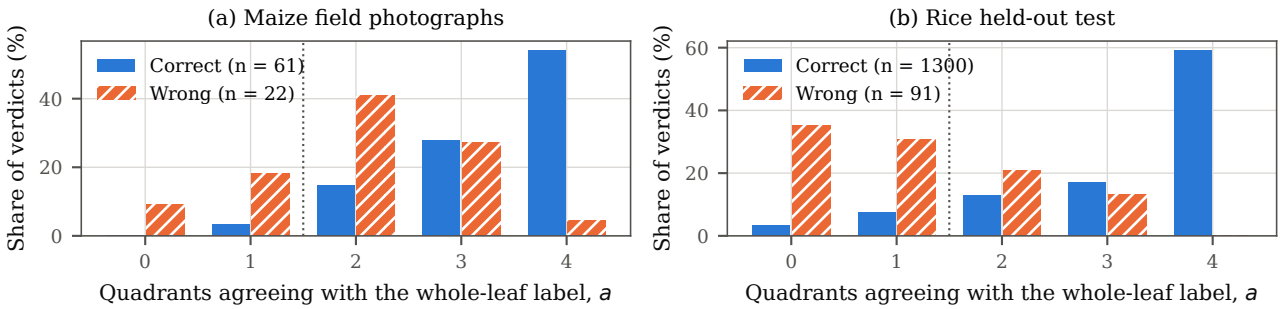


Figure 6: Distribution of the quadrant agreement count a for correct and wrong verdicts. Bars left of the dotted line ($a < 2$) are flagged by the deployed gate.

cross-validated logistic-regression gate did not beat MSP on either set: AURC 0.128 against 0.164 on field photographs (difference −0.036, −0.129 to +0.052) and 0.0054 against 0.0055 on rice (−0.0005 to +0.0005). On the field set it was also worse than quadrant agreement alone, which is unsurprising with 22 errors to learn from. Table X gives the conformal results. On rice, a target selective error of $\alpha = 0.05$ was met by abstaining on about 5% of images (mean coverage 94.7% with MSP, 95.3% with the learned gate; mean test risk about 4.0%). Test risk exceeded α in 4 and 3 of the 20 splits, above the 2 expected on average but within binomial variation ($P[X \geq 4] \approx 0.13$ for $X \sim \text{Bin}(20, 0.1)$). On the field photographs no threshold could be certified: with about 40 calibration images and a 26.5% error rate, the upper confidence bound never fell below α, and coverage was zero in every split but one. Split-conformal sets met their nominal coverage on both sets, as theory promises under exchangeability within a set. On rice they were nearly all singletons; on field photographs the 95% sets averaged 5.0 of 6 labels with only 8% singletons—correct, but a fair summary of how little the maize model knows about these photographs.

D Which Model Choices Hold Up?

Retrained rice model. Across the three seeds the repository script gave 92.48 ± 2.96% accuracy and macro F1 0.863 ± 0.032 (Table XI). Part of the spread comes from the split rather than the training, since each seed yields a different test set. Replayed through the application code, the seed-42 TFLite model classified 1,300 of 1,391 test images correctly, 93.46% (92.0–94.6%); the script's own evaluation counted 1,298, a two-image difference we attribute to the application's image-loading path. Weeds and healthy leaves were almost always right (recall 0.992–1.000; Fig. 3(a)). Leaf scald was weakest at 0.644: 23 of its 101 images were called bacterial leaf blight, a plausible confusion since both produce elongated lesions from the tip or margin, and bacterial leaf blight in turn lost 13 images to leaf blast.

The currently deployed rice model. The rice model now in the application classified 1,373 of the same 1,391 images correctly, 98.71% (98.0–99.2%); on 76 images only it was right and on 3 only the retrained model (exact McNemar $p < 10^{-18}$). We do not read this as evidence that it is better. Its training data and split are not recorded; if, as seems likely, it was trained on the same public collection with a file-level split, many of these test images or their augmented siblings were in its training set. It classified only one of four labelled rice photographs from outside

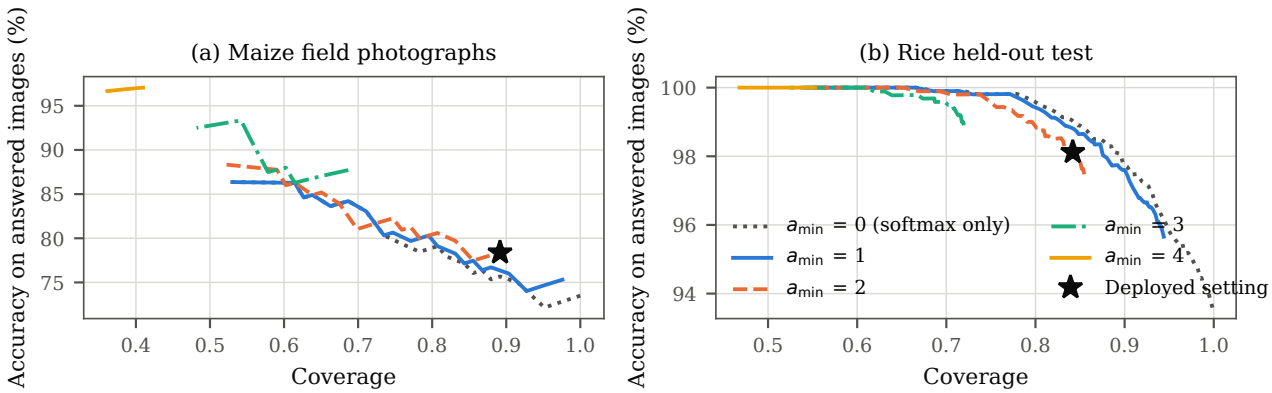


Figure 7: Accuracy on answered images against coverage as the softmax threshold τ varies from 0.30 to 0.99, for each minimum quadrant agreement $a_{\min}$ ($a_{\min} = 0$ is a softmax-only threshold). The star marks the deployed setting.

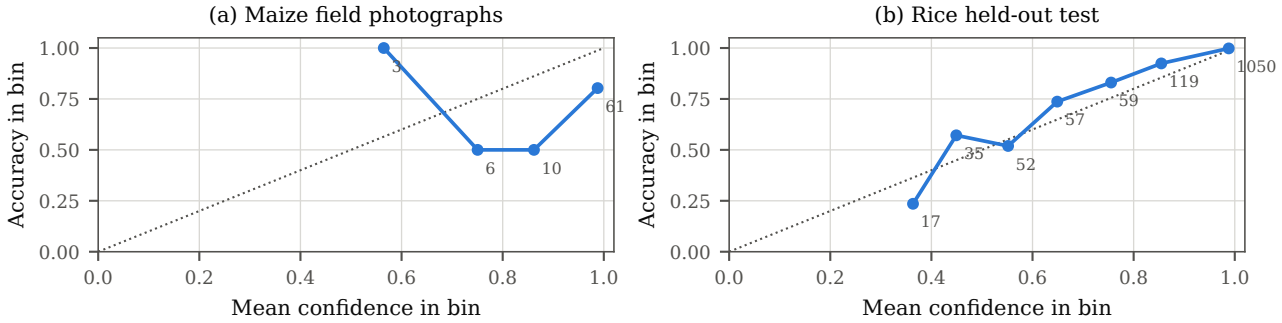


Figure 8: Reliability diagrams of the raw softmax score (10 bins; bins with fewer than three images omitted; numbers give images per bin). Points below the diagonal are overconfident.

the collection correctly (Appendix A). The breakdown by file origin fits that reading (Table XII): its lead is largest on pre-augmented copies (98.8% vs. 87.3%), the files most likely to have siblings in its training set, while the two models are level on the remaining files. On the 40 camera originals it was still ahead, 33 against 29 correct, but with 5 against 1 discordant images the difference is not significant ($p = 0.22$). The same caution applies, more weakly, to the retrained model: if the hash missed some augmented siblings, its 93.5% is also optimistic, and 72.5% (57.2–83.9%) on 40 camera originals hints at where accuracy on unedited photographs may lie.

Backbones. Table XIII and Fig. 9 compare three backbones trained three times each on the same rice split. ConvNeXt-Tiny was the most accurate, 94.08 ± 0.47% with macro F1 0.909, and every ConvNeXt run beat every run of the other two. It is also 4.7 times larger than EfficientNetV2-B0, 3.4 times slower on the CPU and took about 30 min per run against 4–5 min. MobileNetV3-Large, the smallest and fastest, was slightly more accurate than EfficientNetV2-B0 (91.90% vs. 91.25%) and the most consistent across seeds (SD 0.17 points); the 0.65-point gap lies within the spread of the EfficientNetV2-B0 runs, so we do not rank them.

Two cautions apply. First, the backbone study's EfficientNetV2-B0 (91.25%) is 2.2 points below the repository script's EfficientNetV2-B0 on the same seed-42 split (93.46% replayed). The two runs differ in training code, schedule, class weighting and precision policy, and we did not isolate which difference accounts for the gap; measured against the stronger 93.46% model, ConvNeXt-Tiny gains about 0.6 points rather than 2.8. Second, these are in-distribution numbers; the rice field set was missing, and a backbone that ranks first on curated images need not rank first on field photographs [8].

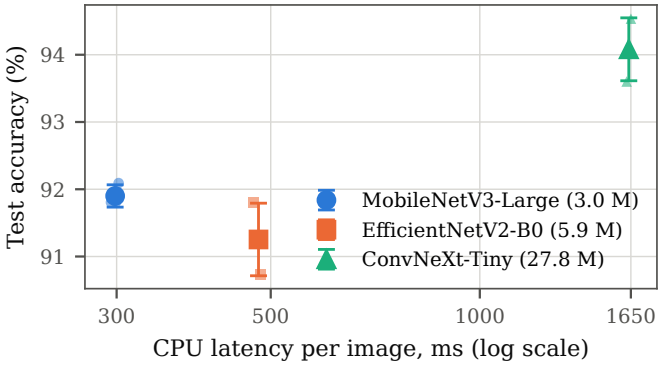


Figure 9: Accuracy against CPU latency for the three rice backbones. Small markers are individual seeds; large markers show the mean with ±1 SD.

Table X: Risk Control and Conformal Prediction Sets (20 Random Splits)

	Maize field	Rice test
Risk control, $\alpha = 0.05$, $\delta = 0.1$ (MSP / learned)		
Mean coverage	0.0 / 0.0	94.7% / 95.3%
Mean test risk	–	4.01% / 3.95%
Splits with risk $> \alpha$	–	4 / 3
Risk control, $\alpha = 0.10$, $\delta = 0.1$ (MSP / learned)		
Mean coverage	0.0 / 2.0%	99.9% / 99.8%
Splits with risk $> \alpha$	0 / 1	0 / 0
Prediction sets, nominal 90%		
Empirical coverage	92.6%	90.3%
Mean set size	2.44	0.94
Singleton sets	31.6%	93.5%
Prediction sets, nominal 95%		
Empirical coverage	97.4%	95.0%
Mean set size	5.01	1.05
Singleton sets	7.9%	94.2%

Table XI: Rice Retraining With the Repository Script (EfficientNetV2-B0)

Seed	Test n	Accuracy (%)	95% CI	Macro F1
42	1,391	93.31	91.9–94.5	0.894
7	1,285	89.18	87.4–90.8	0.831
2024	2,682	94.93	94.0–95.7	0.865
Mean ± SD		92.48 ± 2.96		0.863 ± 0.032

Few-shot field adaptation. Adding field photographs to a linear head did not help (Fig. 10). With laboratory images only, the frozen-feature head reached 68.7% on the field photographs; with 5, 10 and 20 field photographs per class it reached $66.2 \pm 2.1\%$, $69.8 \pm 4.3\%$ and $68.6 \pm 1.6\%$. None differs meaningfully from the starting point, and all stay below the 73.5% of the fine-tuned deployed model. A linear head on frozen ImageNet features has little room to adapt, and with only 12 gray-leaf-spot photographs the larger budgets leave few test images for that class. We read the result as "a handful of photographs is not enough for this cheap form of adaptation", not as evidence that field data cannot help.

VLM cross-check. The planned measurement failed. All 60 calls for the maize field sample returned the application's "configured Gemini model is not available to this key" error, and no calls were made for cotton and rice because their field sets were missing. We therefore have no data on agreement rates or on whether agreement predicts correctness. The failure did exercise R5 under realistic conditions: each call reported a specific reason, and the classifier result was produced and recorded for all 83 photographs. The cause is a mismatch between the API key and the model alias, which needs no code change.

Table XII: Rice Test Accuracy by File Origin: Retrained vs. Deployed Model

File origin	n	Retrained (%)	Deployed (%)
Camera originals (`IMG_*`)	40	72.5	82.5
Pre-augmented copies (`aug_*`)	568	87.3	98.8
Other (incl. weeds)	783	99.0	99.5
All	1,391	93.5	98.7

Table XIII: Rice Backbone Comparison (Same Split, $n =$ 1,391, Three Seeds)

Backbone	Acc. (%)	Macro F1	Params	CPU ms*	Train
MobileNetV3-L	91.90 ± 0.17	0.874 ± 0.004	3.0 M	294	3.8 min
EffNetV2-B0	91.25 ± 0.54	0.863 ± 0.009	5.9 M	483	4.6 min
ConvNeXt-T	$\mathbf{94.08 \pm 0.47}$	$\mathbf{0.909 \pm 0.008}$	27.8 M	1,650	29.7 min

*Median single-image latency on the Colab CPU under `mixed_float16` (seed-42 model); relative only. Train time per run on a T4 GPU, mean of three seeds.

VI Discussion

A What the Evidence Supports

Table XIV states, for each component, what has been measured and what has not. Three findings carry most of the weight.

A file-level split is not a test for these datasets. Without screening, 297 of 380 "field" maize photographs would have been training images and the reported field accuracy would have sat near the laboratory figure. The rice results show the same problem from the other side: a model of unknown provenance scores 98.7% on a split it has probably seen, and its lead is concentrated on pre-augmented copies. Public plant collections that merge overlapping sources and ship augmented files need group-aware splits and screened field sets before any number is reported.

Review rules based on known failure, not on confidence, are what stop silent crop errors. The router misroutes about half of all rice leaves, three quarters of them above its confidence threshold. A confidence-only rule would have let 248 of 684 rice leaves through to the maize model; the crop-based rule let none through. The two cotton false accepts show the remaining hole.

Part consistency is a useful error signal under shift. On field photographs, ranking verdicts by how many quadrants agree with the whole-leaf label cut the AURC of MSP by 44%, and it was the only one of seven scores whose interval excluded zero; in distribution it was slightly worse. One explanation, which we have not tested, is that under shift the network's confidence is driven by cues unrelated to the lesion, such as background, lighting or camera, and so stays high on wrong answers. The quadrant test, by contrast, asks whether the same label is supported by different parts of the leaf, which a shortcut is less likely to satisfy everywhere [20]. An ablation on background-masked field photographs would test this. In distribution the softmax score already carries most of the information.

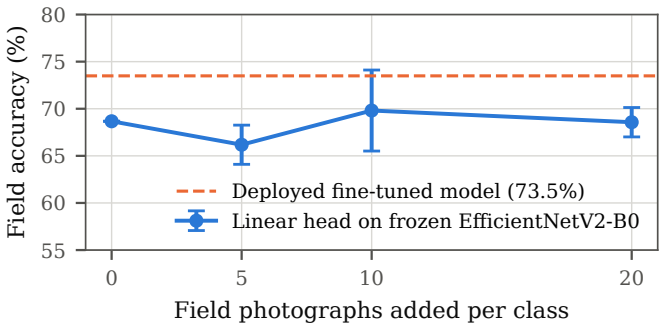


Figure 10: Few-shot adaptation: a linear head on frozen EfficientNetV2-B0 features, trained on 900 laboratory images plus 0–20 field photographs per class (mean ± SD over five draws), never reaches the deployed fine-tuned model (dashed).

B What Did Not Work

The deployed verdict gate removes most in-distribution errors, but a plain softmax threshold at the same coverage removes slightly more; the hand-set combination is not an improvement over the baseline it was meant to replace. The learned gate did not beat MSP on either set, and with 22 field errors it could not be expected to. Few-shot adaptation with a linear head gave nothing. Each points to the same constraint: the decisions that matter are made under field shift, and we have 83 field photographs of one crop to tune them on.

C Implications for Deployed Diagnosis Tools

The results suggest changes that we list without having made them.

1. Keep the crop-review gate and add a review band for cotton suggestions (for example below 97%) until the router is retrained with rice images.

2. Replace the fixed rule (2) by a ranking on quadrant agreement and softmax score, and set its threshold on a labelled field set by risk control, so that the review rate corresponds to a stated error target.

3. Fit any temperature on field photographs; a laboratory temperature would sharpen predictions that are already overconfident.

4. Deploy neither rice model on its in-distribution score; compare both on a rice field set. For a phone build, train MobileNetV3-Large; for server inference, ConvNeXt-Tiny is worth evaluating against the repository-script model.

5. Withhold treatment plans for results marked for review in the classifier-only mode as well.

D Comparison With Published Results

Our laboratory figures sit in the range reported for curated data, below the 99% of PlantVillage studies [4], [5] and close to the 95.99% that Haque et al. obtained on maize photographs from Indian experimental fields [22]. The cotton model reaches 94.43% on five CottonWeedID15 classes while sharing capacity with leaf conditions; Chen et al. report F1 up to 98.93% on all 15 classes with dedicated weed classifiers [32], so the joint model probably gives up some weed accuracy. The 21.5-point maize drop is consistent in direction with the laboratory–field gaps in [4], [5], [19]. None of these comparisons is direct.

E Threats to Validity

Internal. The cotton and maize models were trained once; only rice was trained with three seeds, and its test set changes with the seed. For maize, the validation split is a file-level split reused across runs, and the notebook drew it through the same generator as the training data, so validation images were randomly augmented. Our duplicate check uses one hash and one threshold and does not catch rotated or cropped copies, so residual leakage in the rice test split is possible, and at least 50 mixed-label hash groups are unaudited. The evaluation used the decision functions at commit ba20fa7.

External. The only field set is 83 PlantDoc maize photographs collected from the web, in three disease classes, without healthy leaves or weeds. They are not photographs taken by farmers in India with ordinary phones. The weed species come from US cotton fields [32] and a Brazilian soybean dataset [30].

Construct. Accuracy on single, centred images does not measure whether a farmer receives a correct and safe recommendation; the VLM cross-check and the advice are not covered by any number here.

Statistical conclusion. We compared seven confidence scores and report the one difference that stayed significant after Bonferroni correction (two-sided $p \approx 0.006$); the field sample is small, the interval is wide, and replication on other field sets is needed before the finding is relied upon.

F Protocol for the Field Study

Table XV specifies the study that would settle the open questions. Precision sets the sample size: estimating an accuracy near 75% to within ±5 points at 95% confidence needs about $1.96^2 \cdot 0.75 \cdot 0.25 / 0.05^2 \approx 288$ photographs per crop, and risk control at $\alpha = 0.05$ becomes usable only once a calibration set contains a few hundred images.

G Limitations

The largest limitation is the coverage of the field evaluation: cotton and rice have no field results, the rice

Table XIV: Evidence Status of Each Component After This Audit

Component	Status	Evidence	Missing
Cotton classifier	validated (lab)	95.57% on 790 test images; TFLite 95.82%	field set (download failed)
Maize classifier	validated (lab, field)	94.98% lab leaves; 73.5% on 83 screened field photographs	larger field set
Rice classifier (retrained)	validated (lab)	93.46% held out, 3 seeds; 72.5% on 40 camera originals	field set
Rice classifier (deployed)	not validated	98.7% on a split it has probably seen; 1 of 4 external photographs	provenance
Crop router	evaluated: weak	51.2% correct on rice leaves; 94.0% on maize field	retraining
Crop-review gate	evaluated: works	0/684 wrong-crop auto-accepts on rice; 2/83 via cotton	cotton review band
Whole-leaf rule vs. quadrant vote	evaluated: better	+8.1 points on rice ($p < 10^{-17}$); tie on maize field	–
Verdict gate (τ, $a_{\min}$)	evaluated: mixed	catches 76% of rice errors; not better than MSP at matched coverage	field-tuned thresholds
Quadrant agreement as a score	evaluated: helps under shift	AURC 0.092 vs. 0.164 on field; worse in distribution	replication
Learned gate	evaluated: null	no significant AURC gain on either set	more field errors
Risk control / conformal sets	evaluated	valid on rice; nothing certifiable on 83 field photographs	field calibration data
Few-shot adaptation	evaluated: null	68.7% → 68.6% with 20 photographs per class	fine-tuning, more data
Blind VLM cross-check	not evaluated	all calls failed (model unavailable to key)	re-run
Prescription safety	not evaluated	prompt constraints only	agronomist review

Table XV: Proposed Field-Evaluation Protocol

Item	Specification
Images	≥ 300 phone photographs per crop, ≥ 30 per class, healthy leaves and weeds included, from at least three farms
Labels	two agronomists label independently; Cohen's κ reported; disagreements adjudicated
Leakage	hash screening against all training data; farm-level split between calibration and test halves
Primary metrics	accuracy and macro F1 with Wilson intervals; router accuracy; wrong-crop auto-accept rate
Selective metrics	AURC per score; accuracy at matched coverage; risk control at $\alpha \in \{0.05, 0.10\}$
VLM	agreement categories against correctness; plan-withholding rate
Advice	agronomist rating of 50 generated plans for correctness and safety
Pass criteria	wrong-crop auto-accept $< 1\%$; certified selective error $\leq 10\%$ at coverage $\geq 60\%$

backbones have no field ranking, and the planned leave-one-crop-out experiment could not run. The VLM run failed, so the cross-check that gates treatment advice is unmeasured. Field boundaries are drawn by hand, soil moisture is a model estimate grouped into fixed bands that ignore soil texture, and saved fields last only for the browser session. The interface is English-only and needs a connection; an offline phone build, visual explanations such as Grad-CAM [73] and Indian-language interfaces are planned.

VII Conclusion

Agri-Doctor Pro diagnoses diseases and weeds in maize, rice and cotton with a crop router, crop-specific classifiers, review gates and a blind VLM reading. Auditing its decision code under leakage control changed what we can claim. Leakage screening removed 78% of the field test set; on what remained, the maize model lost 21.5 points against laboratory leaves and made most of its errors with high confidence. The router was close to chance on rice leaves, yet a review rule based on known confusion kept every misrouted rice leaf from being diagnosed silently. The deployed verdict gate was no better than a softmax threshold, while quadrant agreement, a by-product of the discarded decision rule, was the best available signal of field errors. Conformal methods kept their promises but, on 83 photographs, promised little.

The remaining questions are field questions. With cotton and rice field photographs, a working VLM configuration and a few hundred expert-labelled images per crop, the same replay protocol would measure the cross-check, set review thresholds by risk control rather than by hand, and decide which rice model to deploy.

A Qualitative Regression Photographs

Twelve Wikimedia Commons photographs used by the regression tests are too few to estimate accuracy but serve as a case study. Two cotton photographs were routed to cotton and accepted. One maize common-rust photograph was diagnosed correctly with all four quadrants agreeing. One maize blight photograph kept its correct whole-leaf label but had fewer than two agreeing quadrants and was marked for review, where the old rule would have reported a quadrant-level disease. A healthy maize leaf was diagnosed correctly. Of the seven rice photographs, routing is documented for five leaves: three went to maize at 92–97% confidence, and all seven were flagged by the crop gate. Four carry disease labels (two leaf blast, brown spot, sheath blight), and the deployed rice model labelled one of them correctly; the remaining three show whole plants.

B VLM Assessment and Prescription Prompts

The assessment prompt asks Gemini to act as a plant pathologist examining one photograph, to judge only from what is visible, and states that no other system's diagnosis is provided. It asks for the fields described in Section III with a glossary of the 22 labels, instructs the model to report low certainty rather than guess, and requires nutrient deficiency and mechanical or herbicide damage among the alternatives where relevant. The prescription prompt supplies the crop, the detected issue, location, weather, soil type and moisture, the whole-leaf reading, the top three candidates per quadrant, any review reasons and, when available, the VLM assessment and comparison category. It requests five sections: diagnosis summary, field-conditions check, weather check for spraying, action plan and organic alternative. It forbids presenting the classification as a confirmed laboratory diagnosis, giving pesticide doses, mixing or application rates, or guaranteeing recovery; it requires a recommendation that a local extension officer verify the diagnosis before any chemical treatment.

C Reproducibility and Data Availability

The application, the rice training script and the tests are in the public repository [12]; the evaluation used commit ba20fa7. The evaluation notebook (`Agri_Doctor_full_evaluation.ipynb`, with its embedded `exp_lib.py`), the per-image records, the summary files and the independent re-analysis and figure scripts are [to be added to the repository before submission]. Each record holds the true label, all class probabilities, the two flipped-copy probability vectors, quadrant agreement, review flag, the old rule's prediction, router crop and confidence, and the file path. Two notebook bugs were fixed during the run: `model.compile()` had been called with positional arguments, which Keras 3 maps to `loss_weights`, and ConvNeXt CPU latency failed because XLA-compiled variables lived on the GPU and is now measured on a CPU copy. Recomputing from the records with a separate script reproduced every accuracy, interval, McNemar test, router count, gate count and AURC reported here. The paired-difference intervals for AURC, the matched-coverage intervals and the calibration spread in Table IX come from that script (seed 0), and the per-score AURC intervals in Table VII from the notebook. Gemini calls use the `gemini-2.5-flash` alias, which may point to different snapshots over time. The laboratory datasets [66], [67], [68], [32], [30] and field datasets [6], [26], [27] are public.

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